Jan-Ole Fischer

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EDUCATION

PhD in Statistics 2023 – 2026
Bielefeld University, Bielefeld, Germany
Fast and scalable inference in richly parameterised latent Markov models
Grade: summa cum laude; supervisor: Prof. Dr. Roland Langrock

Research Stay March–April 2025
University of British Columbia, Vancouver, Canada

MSc Statistical Science 2021 – 2023
Bielefeld University, Bielefeld, Germany
Grade: 1.0 (best possible grade in the German system); graduated first of my class

Preparatory Coursework in Economics (for MSc admission) 2020 – 2021
Bielefeld University, Bielefeld, Germany

BA Business Administration 2017 – 2020
Bielefeld University of Applied Sciences, Bielefeld, Germany
Grade: 1.4

Study Abroad September 2019 – January 2020
Budapest Business School, Budapest, Hungary

EMPLOYMENT

Research Associate, Statistics and Data Analysis Group 2023 – today
Bielefeld University, Bielefeld, Germany

Research Assistant, Statistics and Data Analysis Group 2022 – 2023
Bielefeld University, Bielefeld, Germany

Working Student, Pricing Intelligence 2021 – 2022
CLAAS KGaA mbH, Harsewinkel, Germany

Intern, Market Intelligence 2020
CLAAS KGaA mbH, Harsewinkel, Germany

PUBLICATIONS

Earlier publications appear under my former surname, Koslik.

8. **Fischer, J.-O.**, Dupont, F., Auger-Méthé, M., Marcoux, M., Hussey, N. E., Heckman, N. (2026). Flexible unimodal density estimation in hidden Markov models. *Methods in Ecology and Evolution*, 17(10), 2994–3005.

7. Vienken, M. N., **Fischer, J.-O.**, Langrock, R. (2026). Inference on state occupancy in covariate-driven hidden Markov models. *Methods in Ecology and Evolution*, 17(8), 2436–2448.

6. Bajons, R., **Koslik, J.-O.**, Michels, R., Ötting, M. (2026). PEP: a tackle value measuring the prevention of expected points. *Journal of Quantitative Analysis in Sports*, 22(1), 35–51.

5. **Koslik, J.-O.**, Feldmann, C. C., Mews, S., Michels, R., Langrock, R. (2025). Inference on the state process of periodically inhomogeneous hidden Markov models for animal behavior. *The Annals of Applied Statistics*, 19(4), 2724–2737.

4. Mews, S., **Koslik, J.-O.**, Langrock, R. (2025). How to build your latent Markov model: The role of time and space. *Statistical Modelling*, 25(6), 481–507.

3. **Koslik, J.-O.** (2025). Hidden semi-Markov models with inhomogeneous state dwell-time distributions. *Computational Statistics & Data Analysis*, 209, 108171.

2. Dyck, J., **Koslik, J.-O.**, Sauzet, O. (2025). EgoCor: an R package to fit exponential semi-variograms to model the local spatial correlation structure of health outcomes. *Journal of Open Research Software*, 13(1).

1. Michels, R., **Koslik, J.-O.** (2024). On the combination of data smoothing and Markov-switching models. Letter to the Editor. *Journal of the Royal Statistical Society – Series C: Applied Statistics*, 73(3), 557–560.

PREPRINTS

Fischer, J.-O. Fast and scalable inference in hidden Markov models with Gaussian fields. In revision at the *Journal of Computational and Graphical Statistics*. arXiv:2603.17469.

Koslik, J.-O. Tensor-product interactions in Markov-switching models. In revision at the *Journal of Agricultural, Biological, and Environmental Statistics*. arXiv:2507.01555.

Koslik, J.-O. Efficient smoothness selection for nonparametric Markov-switching models via quasi restricted maximum likelihood. In revision at *Statistics and Computing*. arXiv:2411.11498.

Dupont, F., McClintock, B. T., **Fischer, J.-O.**, Marcoux, M., Hussey, N. E., Auger-Méthé, M. Inferring resource selection and utilization distributions from irregular and error-prone animal tracking data. arXiv:2606.12566.

Michels, R., Bajons, R., **Fischer, J.-O.** Integrating unsupervised and supervised learning for the prediction of defensive schemes in American football. arXiv:2602.10784.

Ammann, K., Adam, T., **Koslik, J.-O.** Non-homogeneous Markov-switching generalized additive models for location, scale, and shape. arXiv:2601.03760.

SOFTWARE

I have developed and participated in developing software packages for the R programming language:

- **LaMa**: fast and convenient likelihood evaluation for latent Markov models like HMMs, HSMMs, SSMs and point processes.
- **RTMBdist**: distributions compatible with automatic differentiation by RTMB.
- **EgoCor**: simple presentation of estimated exponential semi-variograms.

TEACHING

Workshops

Hidden Markov Models for movement, acceleration and other ecological data — an introduction using moveHMM and momentuHMM in R
Three-day online workshop. September 2024

Using hidden Markov models to uncover ecological state dynamics from biologging data
One-day workshop in Tokyo, Japan. March 2024

Lectures

Hidden Markov Models (Master level) Summer Term 2026

Project-based courses

Statistical Consulting (Master level)
Organiser of the course Winter Term 2026/27
Supervision of a student group through a real client project Winter Terms 2023/24 – 2025/26

Tutorials

Bielefeld University 2023 – today

- Mathematics (Bachelor level)
- Multivariate Methods (Bachelor level)
- Generalised Linear Models (Master level)
- Hidden Markov Models (Master level)

Bielefeld University of Applied Sciences 2018 – 2020
- Statistics for Economists (Bachelor level)
- Mathematics for Economists (Bachelor level)

GRANTS AND AWARDS

Best student paper award at the 40th International Workshop on Statistical Modelling 2026

Nomination for the Andreas Handl teaching award, the Faculty's student-nominated prize for staff below professorial rank 2026

Teaching award of the Department of Business Administration and Economics, for courses among the best-evaluated in the department
- "Mathematics II" exercise classes — awarded each summer term 2024–2026 (jointly with J. Dyck in 2024, K. Ammann in 2025 and 2026)
- "Hidden Markov Models" (Master's lecture) — 2026

Mobility grant of the Bielefeld Graduate School in Theoretical Sciences (€4,000) 2024

German public-private scholarship (Deutschlandstipendium) 2019 – 2020 and 2022 – 2023

PRESENTATIONS

Fast and scalable inference in hidden Markov models with Gaussian fields
Talk at the International Workshop on Statistical Modelling (IWSM), University of Oslo, Norway, July 2026

Fast estimation of state-switching correlated random walk models subject to measurement error
Talk at the BioInference Meeting, University of St Andrews, Scotland, June 2026

Efficient estimation of nonparametric Markov-switching models
Talk at the National Centre for Statistical Ecology Meeting, University of St Andrews, Scotland, December 2025

Fast approximate likelihood inference for indirectly observed hidden Markov models
Talk at the Center for Statistics, Bielefeld University, Germany, December 2025

Hidden semi-Markov models with inhomogeneous state dwell-time distributions
Invited talk, MIAT Seminar, INRAE, Toulouse, France, November 2025

Efficient estimation of nonparametric Markov-switching models
Talk at the Statistical Week 2025, Wiesbaden, Germany, September 2025

Latent Markov models in ecology: a unifying perspective
Invited talk at the Nordic-Baltic Biometric Conference (NBBC), Oslo, Norway, June 2025

Efficient smoothness selection for nonparametric Markov-switching models via quasi restricted maximum likelihood
Invited talk, University of St Andrews, Scotland, February 2025

Efficient smoothness selection for nonparametric Markov-switching models via quasi restricted maximum likelihood
Invited talk, University of Göttingen, Germany, September 2024

Exploring hidden dynamics: statistical inference in periodically inhomogeneous hidden Markov models for animal behaviour
Talk at the Royal Statistical Society Conference (RSS), Brighton, UK, September 2024

Inference on the state process of periodically inhomogeneous hidden Markov models for animal behaviour
Talk at the International Statistical Ecology Conference (ISEC), Swansea, Wales, July 2024

Mitigating consequences of the Markov assumption
Poster at the 18th doctoral workshop of the Bielefeld Graduate School of Economics and Management (BiGSEM), Bielefeld University, Germany, December 2023

Mitigating consequences of the Markov assumption
Talk at the Center for Statistics, Bielefeld University, Germany, October 2023

COMMUNITY INVOLVEMENT

Reviewer

- Journal of Computational and Graphical Statistics
- Journal of the Royal Statistical Society Series A
- Journal of the Royal Statistical Society Series C
- Computational Statistics and Data Analysis
- Methods in Ecology and Evolution
- Movement Ecology
- TEST

SKILLS

- **Programming:** R (including Rcpp), C++, Python
- **Tools:** LaTeX, git, SQL
- **Languages:** German (native), English (fluent)

VOLUNTEERING

Member of the student council (Data Science & Statistical Science) 2021–2023
Bielefeld University, Germany

Representing the interests of student assistants (SHK-Rat) 2022
Bielefeld University, Germany

Voluntary service 2016–2017
AWO nursery school Werther, Germany